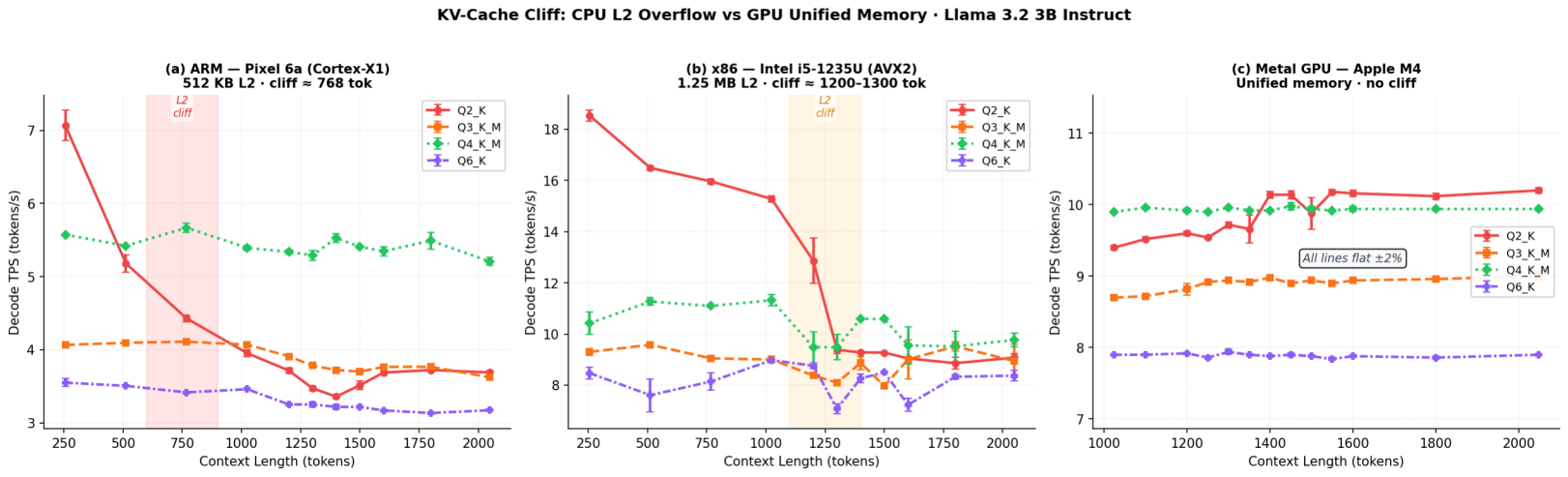
KV-Cache Cliff: CPU L2 Overflow vs GPU Unified Memory · Llama 3.2 3B Instruct
(a) ARM — Pixel 6a (Cortex-X1)
512 KB L2 · cliff ≈ 768 tok
L2 cliff
Q2_K
Q3_K_M
Q4_K_M
Q6_K
Decode TPS (tokens/s)
Context Length (tokens)
(b) x86 — Intel i5-1235U (AVX2)
1.25 MB L2 · cliff ≈ 1200–1300 tok
L2 cliff
Q2_K
Q3_K_M
Q4_K_M
Q6_K
Decode TPS (tokens/s)
Context Length (tokens)
(c) Metal GPU — Apple M4
Unified memory · no cliff
All lines flat ±2%
Q2_K
Q3_K_M
Q4_K_M
Q6_K
Decode TPS (tokens/s)
Context Length (tokens)